

Aptimizer Calculation Reference

Every formula the platform computes, extracted from the source. Appendix to the Validation Protocol.

How to use this

These formulas are transcribed from the code, not from a textbook. Where the code simplifies, the simplification is stated. That makes this document the thing to hand-check against a standard: a difference between this page and the code is a transcription error, and a difference between this page and the IS clause is an engineering error.

Constants live in one place. `iscodes.py` is the single source of truth for every IS and NBC value, and no module carries its own copy. To check a constant, check it there once.

14 modules, 106 formulas and 72 constants follow, in workflow order.

Areas, FAR and FSI

Codes: Project definition

engine.py – tower_metrics, area_metrics

Quantity	Formula as implemented	Note
Tower height	$H = \text{floors} \times \text{floor_height}$	m
Carpet per floor	$\text{sum}(\text{unit_carpet} \times \text{unit_count})$	over the unit mix
Built-up per floor	$(\text{carpet} + \text{balcony}) \times (1 + \text{wall_factor}) + \text{service_core}$	wall_factor default 0.10; service core = corridor + stairs + lifts
Service core	$\text{corridor_w} \times \text{corridor_L} + \text{sum}(\text{stair_w}^2 \times 2.6 \times n) + 4.5 \times n_lifts$	2.6 is the flight-plus-landing factor; 4.5 sq m per lift
Super built-up	$\text{builtup} \times (1 + \text{loading}) + \text{society_amenities}$	loading default 0.25
Ground coverage	$\text{footprint} / \text{plot_area} \times 100$	%
FAR	$\text{total_builtup} / \text{plot_area}$	dimensionless
FSI	$\text{FAR} \times \text{fsi_factor}$	fsi_factor from project config
Open space	$\text{plot_area} - \text{footprint}$	sq m
Density	$\text{units} / \text{plot_area_acres}, \text{occupants} / \text{plot_area_hectare}$	two measures

Constants used

Constant	Value	Source
wall_thickness_factor	0.10	config
common_area_loading	0.25	config
lift area	4.5 sq m each	engine.py
stair factor	2.6	engine.py

Module 1: Structural Load Estimator

Codes: IS 875 Parts 1-3, IS 456:2000

engineering.py – m1_structural_loads

Quantity	Formula as implemented	Note
Slab self-weight	$w_{slab} = 25.0 \times t$	t in m; 25 kN/cu m for RCC
Wall load on slab	$w_{wall} = \gamma_{wall} \times t_{wall} \times h_{wall} \times 0.35$	h_{wall} = floor_height - t, floor min 2.4 m; 0.35 m of wall per sq m of plate
Total dead load	$DL = w_{slab} + finishes + w_{wall}$	kN/sq m
Factored load	$w_u = 1.5 \text{ DL} + 1.5 \text{ LL}$	IS 456 Table 18
Tributary area	$A_t = bay_x \times bay_y$	sq m
Column axial load	$P_u = w_u \times A_t \times floors$	kN, accumulated to the base
Column capacity	$P_u = (0.40 f_{ck} + (0.67 f_y - 0.40 f_{ck}) p) \times A_g$	IS 456 Cl. 39.3, short column; p = steel fraction, min 0.008
Required gross area	$A_g = P_u \times 1000 / (0.40 f_{ck} + (0.67 f_y - 0.40 f_{ck}) p)$	sq mm
Slenderness	$\lambda = l_{unsupported} / \text{least lateral dimension}$	short if $\lambda < 12$, IS 456 Cl. 25.1.2
Minimum eccentricity	$e_{min} = \max(l/500 + D/30, 20 \text{ mm})$	Cl. 39.3 valid only while $e_{min} \leq 0.05 D$
Beam depth	$d = \text{span} / 12$ (simply supported) or $\text{span} / 15$ (continuous)	rounded up to 25 mm; width = $d/2$, min 230 mm
Design wind speed	$V_z = V_b k_1 k_2 k_3$	IS 875-3 Cl. 6.3
Wind pressure	$p_z = 0.6 V_z^2 / 1000$	kN/sq m
Design pressure	$p_d = p_z \times K_d \times K_a \times K_c$	0.9 x 0.9 x 0.9
Lateral wind force	$F = C_f \times A_e \times p_d$	IS 875-3 Cl. 7.4; A_e = face width x height

Constants used

Constant	Value	Source
RCC unit weight	25.0 kN/cu m	IS 875-1
Brick masonry	19.2 kN/cu m	IS 875-1
AAC block	8.0 kN/cu m	IS 875-1
Residential live load	2.0 kN/sq m	IS 875-2
Accessible roof	1.5 kN/sq m	IS 875-2
K_d, K_a, K_c	0.9, 0.9, 0.9	IS 875-3
k_2 by height	1.00 at 10 m to 1.30 at 200 m	IS 875-3 Table 2
Min steel	0.8%	IS 456 Cl. 26.5.3.1

Module 2: Seismic Zone and Base Shear

Codes: IS 1893 (Part 1):2016, IS 13920:2016

engineering.py – m2_seismic

Quantity	Formula as implemented	Note
Fundamental period	$T_a = 0.075 H^{0.75}$ (bare RC frame)	IS 1893 Cl. 7.6.2(a)
Infilled frame period	$T_a = 0.09 H / \sqrt{d}$	Cl. 7.6.2(b); d = base dimension, taken as $\sqrt{\text{footprint}}$
Spectral acceleration, soil I	$S_a/g = 2.5$ for $T < 0.40$; $1.00/T$ for $0.40 \leq T \leq 4$; else 0.25	IS 1893 Fig. 2
Soil II	$S_a/g = 2.5$ for $T < 0.55$; $1.36/T$ for $0.55 \leq T \leq 4$; else 0.34	
Soil III	$S_a/g = 2.5$ for $T < 0.67$; $1.67/T$ for $0.67 \leq T \leq 4$; else 0.42	
Design coefficient	$A_h = (Z \times I \times S_a/g) / (2 R)$	IS 1893 Cl. 6.4.2
Seismic weight	$W = (DL + 0.25 LL) \times \text{built-up area}$	Cl. 7.3.1, 25% live load for $LL \leq 3 \text{ kN/sq m}$
Design base shear	$V_b = A_h \times W$	kN

Constants used

Constant	Value	Source
Zone factor Z	II 0.10, III 0.16, IV 0.24, V 0.36	IS 1893 Table 3
Response R	OMRF 3.0, SMRF 5.0, Shear wall 4.0, Dual 5.0, Flat slab 3.0	Table 9
Importance I	residential 1.0, important 1.2, critical 1.5	Table 8
Soil mapping	hard rock I; gravel, dense sand, stiff clay II; soft clay III	Cl. 6.4.2

Module 3: Foundation Advisor

Codes: IS 6403:1981, IS 1904:1986

engineering.py – m3_foundation

Quantity	Formula as implemented	Note
Service load	$P_{\text{service}} = P_u / 1.5$	<i>unfactored, for bearing</i>
Required area	$A_{\text{req}} = P_{\text{service}} / \text{SBC}$	<i>sq m</i>
Footing side	$B = \sqrt{A_{\text{req}}} + 0.05$	<i>m, square footing with margin</i>
Applied pressure	$q_{\text{applied}} = P_{\text{service}} / B^2$	<i>kN/sq m, against provided area</i>
Utilisation	$q_{\text{applied}} / \text{SBC} \times 100$	<i>%</i>
Minimum depth (Rankine)	$D_f = (q / \gamma) \times ((1 - \sin \phi) / (1 + \sin \phi))^2$	<i>min 0.5 m</i>

Constants used

Constant	Value	Source
Hard rock	SBC 3240 kN/sq m, phi 40, gamma 22.0	<i>IS 1904 indicative</i>
Gravel / dense sand	SBC 440, phi 35/33, gamma 20.0/19.5	
Stiff clay	SBC 180, phi 25, gamma 19.0	
Soft clay	SBC 100, phi 15, gamma 17.5	

Module 4: Concrete Mix Design

Codes: IS 10262:2019, IS 456:2000

takeoff.py – mix_proportions

Quantity	Formula as implemented	Note
Target mean strength	$fck_target = fck + 1.65 s$	<i>s from IS 10262 Table 2</i>
Water-cement ratio	w/c from exposure class, capped by durability	
Cement content	$C = water / (w/c)$, not less than the exposure minimum	<i>kg/cu m</i>
Coarse aggregate volume	$V_ca = V_table + 0.01 \times ((0.50 - w/c) / 0.05)$	<i>IS 10262 Table 5 adjustment</i>
Fine aggregate volume	$V_fa = 1 - V_ca$	
Aggregate volume	$V_agg = 1 - air - C/(3.15 \times 1000) - water/1000$	<i>cu m per cu m</i>
Coarse aggregate mass	$V_agg \times V_ca \times 2.74 \times 1000$	<i>kg</i>
Fine aggregate mass	$V_agg \times V_fa \times 2.65 \times 1000$	<i>kg</i>

Constants used

Constant	Value	Source
Std deviation s	M20/M25 4.0, M30-M40 5.0 N/sq mm	<i>IS 10262 Table 2</i>
Water content	10 mm 208, 20 mm 186, 40 mm 165 litre/cu m	<i>Table 4, 25-50 slump</i>
CA volume	10 mm 0.50, 20 mm 0.62, 40 mm 0.71	<i>Table 5</i>
Specific gravity	cement 3.15, fine 2.65, coarse 2.74	
Exposure moderate	w/c 0.50, min cement 300, cover 30 mm	<i>IS 456 Table 5</i>
Exposure severe	w/c 0.45, min cement 320, cover 45 mm	

Module 5: Water Infrastructure

Codes: IS 1172:1993, NBC 2016 Part 9

engine.py – water_demand; engineering.py – m5_water

Quantity	Formula as implemented	Note
Occupancy	$\text{sum}(\text{units} \times \text{persons per unit type})$	<i>per-type, not flat</i>
Daily demand	$Q = \text{persons} \times (135 + 45 + 15)$	<i>litres/day, domestic + flushing + external</i>
Sewage	$S = 0.80 \ Q$	<i>litres/day</i>
Sump volume	$V_{\text{sump}} = Q + \text{fire reserve}$	<i>fire reserve 25 000 L above 15 m</i>
Sump dimensions	$\text{area} = V/1000/\text{depth}$, then $2b \times b$	<i>depth default 2.5 m</i>
Overhead tank	$V_{\text{oht}} = Q / 3$	<i>one-third of daily demand</i>
STP capacity	$\text{STP} = S / 1000$	<i>KLD</i>

Constants used

Constant	Value	Source
Domestic	135 lpcd	IS 1172
Flushing	45 lpcd	IS 1172
External	15 lpcd	IS 1172
Sewage factor	0.80	NBC Part 9
Fire reserve in sump	25 000 L above 15 m	NBC Part 4
Static fire storage	50 000 L above 15 m	NBC Part 4 Table 7
STP selection	$\text{SAFF} \leq 50$, $\text{SBR} \leq 200$, else MBR KLD	<i>engineering judgment</i>

Module 6: Storm Water and Rainwater Harvesting

Codes: IS 3764, NBC 2016 Part 9

engineering.py – m6_storm_rwh

Quantity	Formula as implemented	Note
Weighted runoff coefficient	$C = (A_{\text{roof}} \times 0.85 + A_{\text{rest}} \times C_{\text{site}}) / A_{\text{plot}}$	
Peak runoff (rational)	$Q = C I A / 360$	cu m/s with I in mm/hr and A in hectares
Drain diameter (Manning)	$D = ((Q n^{4/3}) / (\pi \sqrt{S}))^{3/8}$	rounded up to 50 mm, min 150 mm
Full-bore velocity	$v = Q / (\pi D^2 / 4)$	self-cleansing band 0.6 to 3.0 m/s
Annual harvest	$V = A_{\text{roof}} \times (\text{annual rainfall} / 1000) \times 0.85 \times 1000$	litres/year
Recharge pit volume	$V_{\text{pit}} = A_{\text{roof}} \times (I / 1000)$	one hour of peak rainfall, cu m

Constants used

Constant	Value	Source
RCC roof C	0.85	IS 3764
Paved C	0.80	
Lawn C	0.20	
Mixed site C	0.60	
Manning n	0.013	RCC pipe
Drain slope	1 in 120	
RWH mandatory	plot > 200 sq m	state bye-law

Module 7 and Parking Engine

Codes: NBC 2016 Part 4, SP:21, state bye-laws

parking.py; engineering.py – m7_parking_nbc

Quantity	Formula as implemented	Note
NBC ECS demand	$\text{ECS} = \text{ceil}(\text{super_builtup} / 100)$	NBC basis, 1 ECS per 100 sq m
State band demand	per-unit factor selected by carpet-area band	carpet-band, state-aware; this is the governing route
Accessible bays	$\text{ceil}(\text{ECS} / 50)$	one definition shared with Module 9
Two-wheeler equivalence	1 ECS = 3 two-wheelers	
EV provision	20% of ECS	MoHUA / BEE 2019
Shared pool	per-tower demand met from tower supply, shortfall from site pool	

Constants used

Constant	Value	Source
ECS per built-up	1 per 100 sq m	SP:21
ECS area	23 sq m	incl. circulation
Max ramp	12.5%	NBC
Ramp width	3.6 m one-way, 6.0 m two-way	
Basement headroom	2.4 m	
Aisle at 90 deg	6.0 m	
Accessible	1 per 50 ECS	NBC Part 3

Module 8: Fire and Life Safety

Codes: NBC 2016 Part 4, IS 2190

engineering.py – m8_fire

Quantity	Formula as implemented	Note
Travel distance	<code>check travel <= 30 m</code>	<i>Group A-2 residential, NBC Table 4</i>
Staircases	<code>two required above 24 m, clear width >= 1.5 m</code>	
Refuge floors	<code>every 7th floor above 24 m</code>	<i>list generated by floor number</i>
Fire lift	<code>required above 30 m, car >= 1.1 x 2.1 m</code>	<i>stretcher dimension</i>
Pressurisation	<code>stairwell pressurisation above 15 m</code>	
Extinguishers	<code>n = ceil(floor plate / 200)</code>	<i>per floor, compared against provided</i>
Static water	<code>50 000 L above 15 m, plus 25 000 L reserve in the sump</code>	<i>NBC Table 7</i>

Constants used

Constant	Value	Source
<code>max_travel_m</code>	30.0	<i>NBC Part 4 Table 4</i>
<code>two_stair_height_m</code>	24.0	
<code>refuge_above_m</code>	24.0	
<code>refuge_every_floors</code>	7	
<code>fire_lift_above_m</code>	30.0	
<code>pressurisation_above_m</code>	15.0	
<code>extinguisher_per_sqm</code>	200.0	<i>IS 2190</i>

Module 9: Accessibility

Codes: NBC 2016 Part 3, IS 3534, RPwD Act 2016

engineering.py – m9_accessibility

Quantity	Formula as implemented	Note
Ramp slope	check slope $\leq 8.33\%$ (1:12)	
Door width	clear width ≥ 900 mm	
Corridor	≥ 1200 mm required, 1500 mm preferred	wheelchair passing
Lift car	$\geq 1100 \times 1400$ mm	
Handrails	dual at 760 and 900 mm	
Accessible bays	provided $\geq \text{ceil}(\text{ECS} / 50)$	same basis as Module 7

Constants used

Constant	Value	Source
ramp_slope	1:12	NBC Part 3
door_width_mm	900	
corridor_min_mm	1200	
corridor_pref_mm	1500	
lift_car_mm	1100 x 1400	
handrail_mm	760 and 900	

Quantity Take-off

Codes: IS 456, IS 10262, IS 1200 for measurement

takeoff.py – structural_takeoff, grid_counts, beam_layout

Quantity	Formula as implemented	Note
Column count	from the bay grid over the footprint	<i>grid_counts</i>
Column concrete	$A_{\text{column}} \times \text{storey height} \times \text{column count} \times \text{floors}$	<i>cu m</i>
Beam sections	depth = span/12 or span/15, width = depth/2	<i>min 230 x 300 mm</i>
Beam concrete	$\text{sum}(b \times d \times \text{length})$ over the beam layout	<i>cu m</i>
Slab concrete	footprint $\times$ slab thickness $\times$ floors	<i>cu m</i>
Steel	derived from member volumes and percentage steel	<i>kg; checked kg per cu m</i>
Sanity check	concrete per sq m and steel per cu m against expected bands	<i>flags an implausible take-off rather than reporting it silently</i>

Constants used

Constant	Value	Source
Min column	230 mm	IS 456 Cl. 25.1.1
Beam divisor	12 simply supported, 15 continuous	IS 456 Cl. 23.2.1

Cost Estimation

Codes: CPWD DSR or state SOR for rates

engine.py – boq, cost adders

Quantity	Formula as implemented	Note
Material cost	$\text{quantity} \times (1 + \text{wastage}\%) \times \text{rate}$	<i>wastage per material</i>
Works total	$\text{material} + \text{labour} + \text{equipment}$	
Adders in sequence	each percentage applied to the running total, not to the base	<i>profit is earned on preliminaries and tax is charged on the lot</i>
Order	preliminaries, then overhead and profit, then contingency, then escalation, then GST	
Unit costs	$\text{grand total} / \text{units}$, $\text{grand total} / \text{built-up area}$	<i>INR per flat, INR per sq m</i>

Constants used

Constant	Value	Source
Wastage	concrete 2%, cement 3%, steel 3%, bricks 5%, sand and aggregate 6%, tiles 8%	<i>engine.py</i>
Preliminaries	3.0%	
Overhead and profit	12.0%	
Contingency	5.0%	
Escalation	4.0%	
GST	18.0%	<i>works contract</i>

Programme and CPM

Codes: Standard critical path method

schedule.py – build_activities, run_cpm

Quantity	Formula as implemented	Note
Activity duration	$D = \text{ceil}(\text{quantity} / (\text{daily output} \times \text{crew}))$	<i>minimum one day</i>
Forward pass	$ES = \max(\text{EF of predecessors}) + \text{lag}; EF = ES + D$	<i>over a work calendar</i>
Backward pass	$LF = \min(\text{LS of successors}) - \text{lag}; LS = LF - D$	
Total float	$TF = LS - ES = LF - EF$	<i>critical when $TF = 0$</i>
Free float	$FF = \min(\text{ES of successors}) - EF$	
Crew efficiency	duration scaled by a crew efficiency multiplier	<i>diminishing returns</i>
Cash flow	cost spread across each activity's working days, aggregated weekly	
Prop removal	slab span and cement type govern the minimum striking time	<i>safety-critical; audited separately</i>

Constants used

Constant	Value	Source
Calendar	working days with holidays	<i>WorkCalendar</i>
Float verification	recomputed and cross-checked	<i>verify_floats</i>

Development Finance

Codes: Standard development appraisal

finance.py – analyse, cash_flow

Quantity	Formula as implemented	Note
Saleable area	derived from the unit mix and the loading factor	sq m
Revenue	saleable area x sale rate	INR
S-curve spread	construction cost split roughly 15 / 70 / 15 across programme thirds	spreading evenly flatters the early cash position
NPV	$\text{sum}(\text{CF}_t / (1 + r)^t)$	monthly discount rate
IRR	r where NPV = 0, found by bisection	bounds -90% to +100% per month
Break-even	first month where cumulative cash flow turns positive	
Peak funding	the most negative cumulative cash position	INR

Constants used

Constant	Value	Source
S-curve	15 / 70 / 15	finance.py
IRR bounds	-0.90 to +1.00 monthly	bisection

Known Simplifications

Every model simplifies. Stating which simplifications are deliberate is what separates a feasibility tool from one that pretends to be a design office. Declare these before an examiner finds them.

Square-plan wind assumption

The frontal face width is taken as $\sqrt{\text{footprint}}$, so the building is treated as square in plan for wind. A long thin block has a larger windward face than this gives, and the lateral force is under-stated accordingly.

Wall load distribution

Wall load on the slab uses 0.35 m of wall per square metre of floor plate, a typical residential figure. A plan with unusually dense or sparse partitions departs from it.

Short-column sizing

Columns are sized by the IS 456 Cl. 39.3 short-column expression. The code checks slenderness and minimum eccentricity and raises a critical warning when the expression does not apply, but it does not then redesign for combined axial load and moment.

Uniform tributary area

Every column takes bay_x by bay_y. Edge and corner columns carry less, interior columns near openings more. This is a preliminary sizing basis, not an analysis.

Wind force coefficient table

The Cf table carries an explicit unverified flag in the code and raises a warning until it is checked against a controlled copy of IS 875 (Part 3) Table 26. Verify it before quoting any lateral force.

Foundation depth

Minimum depth uses the Rankine expression. Real founding depth is decided by the geotechnical report, water table and adjacent construction, none of which the platform holds.